

Task 10: Baseline Machine Learning Theory Document

Implementation Theory, Concepts, and Q&A

Part 1: Core Machine Learning Concepts

In this task, the machine learning workflow is divided into supervised and unsupervised learning. Supervised learning maps features (the independent variables, or matrix X) to a known target (the dependent variable, or vector y). This includes Regression (predicting a continuous value, optimizing residuals and loss) and Classification (predicting discrete class labels using decision thresholds). Unsupervised learning, like Clustering, groups data based on feature similarity without any given targets, utilizing centroids and distance metrics.

Model optimization heavily relies on Gradient Descent, where parameters are iteratively updated to minimize a loss function. The learning rate determines the step size of these updates, and feature scaling is critical because unscaled features cause distorted gradients, hindering convergence.

To ensure models generalize well to new data, datasets are split into Train, Validation, and Test sets. The model learns on the training set, is tuned on the validation set, and is finally evaluated on the unseen test set. This strict separation prevents Data Leakage, ensuring that no target or test-set information inadvertently influences the training phase.

All models must be evaluated against Baseline Predictors (e.g., predicting the mean for regression, or the majority class for classification). A model is only valuable if it significantly outperforms these trivial baselines.

Part 2: Theory Questions & Answers

1. What is the difference between regression, classification, and clustering?

Regression predicts a continuous numerical target. Classification predicts a discrete categorical class label. Clustering is an unsupervised technique that groups data points based on feature similarity without predefined labels.

2. What are features and targets?

Features (X) are the independent input variables (e.g., sensor readings) used to make predictions. The target (y) is the dependent output variable the model aims to predict.

3. What is the difference between supervised and unsupervised learning?

Supervised learning trains on labeled data (where the target is known), mapping features to targets. Unsupervised learning trains on unlabeled data, finding hidden patterns or groupings without a target variable.

4. What is a baseline predictor?

A baseline predictor is a simple, naive model used as a reference point. For regression, it often predicts the

Task 10: Baseline Machine Learning Theory Document

Implementation Theory, Concepts, and Q&A

mean of the training target. For classification, it predicts the majority class.

5. Why is a model weak if it does not beat the baseline?

If a model performs worse than a naive baseline, it means the algorithm failed to learn any meaningful mathematical relationship from the features.

6. Why do we need train, validation, and test sets?

The training set is used to fit model parameters. The validation set is used to tune hyperparameters and prevent overfitting during development. The test set is a completely unseen holdout used only once to evaluate final real-world performance.

7. Why should the test set not be used repeatedly during model selection?

Repeatedly evaluating on the test set causes information leakage (overfitting to the test set). The model's hyperparameters become biased toward the test data, ruining its ability to provide an unbiased performance estimate.

8. What is gradient descent and what role does the learning rate play?

Gradient descent is an iterative optimization algorithm used to minimize the model's loss function by updating parameters. The learning rate determines the step size taken in the direction of the steepest descent.

9. What happens if the learning rate is too high or too low?

If too high, the algorithm takes oversized steps, causing the loss to overshoot the minimum or diverge to infinity. If too low, it takes tiny steps, making training painfully slow and risking getting stuck in local minima.

10. What is the difference between MAE, RMSE, and R-squared?

MAE (Mean Absolute Error) is the average absolute difference between actuals and predictions. RMSE (Root Mean Squared Error) penalizes larger errors more heavily because errors are squared before averaging. R-squared explains the proportion of target variance predictable by the features.

11. What is the difference between accuracy, precision, recall, and F1-score?

Accuracy is the total correct predictions over all predictions. Precision measures how many predicted positives were actually positive. Recall measures how many actual positives the model successfully identified. F1-score is the harmonic mean of precision and recall.

12. Why is accuracy misleading for imbalanced classification?

If a dataset is 99% Class A and 1% Class B, a naive model predicting only Class A will have 99% accuracy but entirely fail to detect Class B.

13. What does a confusion matrix show?

It shows a grid of True Positives, True Negatives, False Positives, and False Negatives, revealing exactly

Task 10: Baseline Machine Learning Theory Document

Implementation Theory, Concepts, and Q&A

what types of errors the classification model is making.

14. What is inertia in KMeans?

Inertia measures the sum of squared distances from each data point to its assigned cluster centroid. It represents cluster compactness (lower is tighter).

15. What is silhouette score?

Silhouette score measures how similar a point is to its own cluster (cohesion) compared to other clusters (separation). It ranges from -1 to 1; higher values indicate well-defined clusters.

16. What is data leakage? Give three examples relevant to this task.

Data leakage is the unintentional introduction of information from outside the training dataset into the model.

- Target Leakage: Using CO(GT) as a feature to predict 'high_pollution' (since it was derived from CO(GT)).
- Preprocessing Leakage: Standardizing the entire dataset before the train-test split.
- Test-set Leakage: Tuning the learning rate based on repeated evaluations of the test set.